Figure 16.14 A few steps of gradient descent: notice the path made by concatenating the gradient vectors after each update; larger vectors mean the gradient was larger (and the curve slope steeper), so in turn Δx the step update for our variable, also will be larger.

As mentioned, we can stop when we reach a minimum: one of the most important results in calculus is Fermat's theorem[12], that proves a point in the domain of a derivable function is a minimum, or a maximum, if and only if the derivative of that function is zero at that point; therefore, we can just check that all the partial derivatives are zero (or, more realistically, that their value is below some precision).

By using the gradient of a function f to decide how much (and in what direction) we should move, we naturally take big steps when f changes fast, and small steps when f changes slowly: transferred to our marble-race example, the marble would travel fast in a steep, straight section, while we would need to be careful when a turn is near, to avoid going off-track.